

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA1610 **COURSE NAME:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

QUESTIONS: 05

Total Marks:50

Annexure A

Data Interpretation

<i>Criteria</i>	Understanding of Data (2.5)	Analysis (2.5)	Application of Concepts (2)	Conclusion (1.5)	Clarity (1)	<i>Total(10)</i>
<i>Weightage</i>	25%	25%	20%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I _G.HimaBindu(192511377) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G.Hima Bindu

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Annexure A

Q1. Dataset: Hospital Patient Segmentation

Patient ID Age Monthly Medical Expense (₹) Health Risk Score

P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K SSE

1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

1. Given Data and SSE Analysis

K (No. of Clusters)	SSE	Reduction in SSE	Interpretation
1	600	—	All patients in one cluster
2	350	250	Large improvement
3	220	130	Significant improvement
4	200	20	Small improvement
5	190	10	Very small improvement

2. Elbow Point

The SSE decreases rapidly from **K = 1 to K = 3**. After K = 3, the decrease becomes very small.

Therefore:

3. Patient Segmentation

Cluster	Patient Examples	Expense Level	Risk Level	Healthcare Decision
Cluster 1	P1, P4	Low	High	Preventive care, regular monitoring
Cluster 2	P3	Moderate/High	Very High	Specialized treatment and frequent monitoring
Cluster 3	P2, P5	High	Low	Cost management and routine follow-up

4. Impact on Healthcare Decision-Making

Area	Benefit of 3 Clusters
Resource Allocation	Allocates doctors, beds and medicines according to patient needs
Risk Management	Identifies high-risk patients for priority care
Personalized Treatment	Allows different treatment plans for different patient groups
Cost Management	Identifies high-expense patients and supports cost control
Preventive Care	Helps identify patients who need early intervention
Healthcare Planning	Helps hospitals plan staff, facilities and medical resources

5. Why K = 3 is Preferred

For $K = 3$, $SSE = 220$. Increasing the clusters to $K = 4$ reduces SSE only to 200, an improvement of just 20. Increasing to $K = 5$ reduces it further to 190, an improvement of only 10.

Thus, choosing $K = 4$ or $K = 5$ increases complexity without providing significant improvement.

Q2. Dataset: E-Commerce Order Analysis

Order ID Product Category Order Value (₹)

O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behavior.

1. Introduction

Hierarchical clustering is an unsupervised machine learning technique that creates a hierarchy of clusters. The results are represented using a dendrogram.

For this e-commerce dataset, orders are grouped based on:

- Product Category
- Order Value (₹)

The dendrogram shows which orders are most similar and at what distance they are combined.

2. Given Dataset

Order ID	Product Category	Order Value (₹)
O1	Electronics	55,000
O2	Clothing	25,000
O3	Electronics	57,000
O4	Furniture	30,000
O5	Clothing	27,000

3. Formation of Clusters

In hierarchical clustering, initially each order is considered as an individual cluster. The closest or most similar orders are then merged step-by-step.

Stage	Cluster Formation	Reason
1	O1 and O3	Both are Electronics and have similar values: ₹55,000 and ₹57,000
2	O2 and O5	Both are Clothing and have similar values: ₹25,000 and ₹27,000
3	O4 remains separate initially	Furniture category and ₹30,000 value
4	Clothing/Furniture group can merge	O2, O5 and O4 have relatively similar lower order values
5	Electronics group merges at a higher distance	Electronics orders have substantially higher values

Thus, the dendrogram would show O1 and O3 joining at a low distance, while O2 and O5 also join at a low distance.

4. Appropriate Number of Clusters

A reasonable choice is **3 clusters**:

Cluster	Orders	Category/Value Pattern	Interpretation
Cluster 1	O1, O3	Electronics, ₹55,000–₹57,000	High-value electronics purchases
Cluster 2	O2, O5	Clothing, ₹25,000–₹27,000	Similar clothing purchases
Cluster 3	O4	Furniture, ₹30,000	Distinct furniture purchase

The exact number depends on the distance threshold at which the dendrogram is cut. For this small dataset, 3 clusters provide an interpretable segmentation.

5. Significance of the Clusters

Cluster	Purchasing Behavior	Business Significance
Electronics Cluster	High-value purchases	Target customers with premium electronics offers and accessories
Clothing Cluster	Similar medium-value purchases	Use fashion promotions, discounts and product recommendations
Furniture Cluster	Separate purchase pattern	Offer furniture-specific promotions and related products

Benefits to the Retailer

1. Customer behavior analysis: Identifies groups of orders with similar purchasing patterns.
2. Personalized recommendations: Customers associated with a cluster can receive relevant product suggestions.
3. Marketing strategies: Different promotions can be designed for electronics, clothing and furniture buyers.
4. Pricing decisions: High-value and low-value purchasing groups can be analyzed separately.
5. Inventory planning: The retailer can understand which product categories have different purchasing patterns.
6. Cross-selling: Electronics buyers can be offered accessories, while furniture buyers can be offered related home products.

Q3. Dataset: Mobile App User Behavior

User Daily Usage Time (min) Login Frequency In-App Purchase (₹)

U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

1. Introduction

The company wants to segment mobile app users based on their activity patterns. The important features are:

- Daily Usage Time (minutes)
- Login Frequency
- In-App Purchase (₹)

In this question, probability-based membership values are used to determine how strongly a user belongs to different clusters. This represents soft clustering, where a user can belong to more than one cluster with different probabilities.

2. Given Dataset

User	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
U1	120	18	2,000
U2	60	5	5,000
U3	140	22	1,500
U4	50	4	6,000
U5	130	20	1,800

3. Identification of User Behavior

The users show two major behavior patterns.

User Group	Users	Usage	Login Frequency	Purchase	Behavior
Highly Active Users	U1, U3, U5	High	High	Lower	Frequently use the app
Low Activity / High Spending Users	U2, U4	Low	Low	High	Use the app less but spend more

4. Probability-Based Cluster Membership

In soft clustering, each user is assigned a membership probability for each cluster rather than being assigned exclusively to one cluster.

For example, suppose the algorithm produces the following membership values

User	Active User Cluster	High-Spending Cluster	Interpretation
U1	0.90	0.10	Strongly active user
U2	0.15	0.85	Strongly high-spending user
U3	0.95	0.05	Very active user
U4	0.10	0.90	Strongly high-spending user
U5	0.92	0.08	Strongly active user

The probabilities for each user add up to approximately 1.

Therefore, U1 has a 90% membership probability in the Active User cluster and only 10% in the High-Spending cluster.

These probability values are illustrative because actual membership probabilities require running the HAC/soft-clustering algorithm on the dataset.

5. Interpretation of Membership Values

Membership Probability	Meaning
0.80–1.00	Very strong membership
0.60–0.79	Strong membership
0.40–0.59	Moderate/uncertain membership
0.20–0.39	Weak membership
0.00–0.19	Very weak membership

For example, if a user has:

- Active Cluster = 0.70
- High-Spending Cluster = 0.30

the user is mainly an active user, but also shows some characteristics of the high-spending group.

This is useful because real customers do not always fit perfectly into one category.

6. Soft Clustering vs Hard Clustering

Hard Clustering

In hard clustering, each user belongs to **only one cluster**.

Example:

User	Cluster
U1	Active
U2	High Spending
U3	Active
U4	High Spending
U5	Active

A user cannot simultaneously belong to two clusters.

Soft Clustering

In soft clustering, each user has a **membership probability** for every cluster.

Example:

User	Active Cluster	High-Spending Cluster
U1	0.90	0.10
U2	0.15	0.85
U3	0.95	0.05
U4	0.10	0.90
U5	0.92	0.08

7. Advantages of Soft Clustering

Advantage	Explanation
Flexible segmentation	A user can partially belong to multiple groups.
Handles overlapping behavior	Useful when users show characteristics of different customer types.
Better personalization	Recommendations can be based on the strength of membership.
Identifies uncertain users	Users with similar probabilities can be identified for further analysis.
Improved marketing	Offers can be targeted according to different membership levels.
More realistic	Customer behavior is often complex and cannot always be represented by one category.

Q4. Dataset: Agricultural Crop Monitoring

Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

1. Given Dataset

Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

2. What is Normalization?

Normalization is the process of converting features with different ranges into a common scale. Here, the variables have different numerical ranges:

- Rainfall: 280–520 mm
- Soil fertility: 3–9
- Crop yield: 65–90 tons

If clustering is performed directly using distance-based methods such as K-Means, rainfall can have a greater influence because its numerical values are much larger.

A commonly used Min-Max normalization formula is:

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

This converts values approximately into the range 0 to 1.

3. Clustering Before Normalization

Before normalization, the clustering algorithm uses the original values.

Feature	Approximate Range	Effect on Clustering
Rainfall	280–520	Very high influence
Soil Fertility	3–9	Low influence
Crop Yield	65–90	Moderate influence

Because rainfall has much larger numerical values, the distance calculation may be dominated by rainfall.

4. Clustering After Normalization

After normalization, all three features have comparable scales.

Farm	Rainfall Pattern	Soil Fertility	Yield	Overall Interpretation
F1	Moderate	High	High	Favorable agricultural conditions
F2	High	Low	Medium	High rainfall but poor fertility
F3	Moderate	Medium	High	Balanced conditions
F4	Low	Very High	Very High	Highly favorable conditions
F5	Very High	Very Low	Low	Less favorable conditions

5. Difference Before and After Normalization

Aspect	Before Normalization	After Normalization
Feature scale	Different	Comparable
Dominant feature	Rainfall may dominate	No feature dominates due to scale
Distance calculation	Can be biased	More balanced
Cluster formation	Mainly influenced by large numerical values	Based on overall feature patterns
Interpretation	May be misleading	More meaningful
Agricultural analysis	Less reliable	More reliable

Q5. Dataset: Employee Performance Evaluation

Employee Productivity Score Attendance (%) Skill Assessment Score

E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model Score

K=2 0.48

K=3 0.68

K=4 0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyze the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

1. Given Dataset

Employee	Productivity Score	Attendance (%)	Skill Assessment Score
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

2. Silhouette Score

The Silhouette Score is used to evaluate the quality of clustering.

It measures:

- How similar an employee is to employees in the same cluster.
- How different an employee is from employees in other clusters.

The score generally ranges from:

[
-1 \text{ to } +1
]

Silhouette Score	Interpretation
Close to +1	Very good clustering
Around 0	Overlapping clusters
Negative	Poor clustering or incorrect assignment

A higher silhouette score indicates better-defined and more separated clusters.

3. Given Silhouette Scores

Model	Silhouette Score	Interpretation
K = 2	0.48	Moderate clustering quality
K = 3	0.68	Best clustering quality
K = 4	0.52	Moderate clustering quality

The highest score is:

[
\boxed{0.68}
]

which corresponds to:

[
\boxed{K=3}
]

Therefore, K = 3 is the most suitable number of clusters.

4. Comparison of the Models

K = 2 → Score = 0.48

This indicates moderate separation between two employee groups. However, the score is not the highest.

Two clusters may be too broad to properly distinguish different levels of employee performance.

K = 3 → Score = 0.68

This is the highest silhouette score.

It indicates:

- Good separation between clusters.
- Strong similarity among employees within the same cluster.
- Better distinction between employee performance levels.

Therefore, K = 3 provides the best representation of the employee data.

K = 4 → Score = 0.52

Although four clusters can provide more detailed segmentation, the score decreases from 0.68 to 0.52.

This suggests that some clusters may become too small or overlap with one another.

5. Possible Employee Segmentation with $K = 3$

Based on productivity, attendance and skill scores, three meaningful groups can be interpreted as:

Cluster	Employee Examples	Performance Level	Interpretation
Cluster 1	E1, E5	High	Highly productive, skilled and regular employees
Cluster 2	E3	Moderate/High	Good performance with scope for improvement
Cluster 3	E2, E4	Low/Moderate	Lower productivity, attendance and skill scores

6. Contribution to Employee Performance Evaluation

Selecting $K = 3$ helps the company evaluate employees more effectively.

Area	Benefit
High performers	Can be rewarded, promoted or given leadership responsibilities
Moderate performers	Can receive training and development programs
Low performers	Can be provided mentoring and performance improvement plans
Training	Skill gaps can be identified for individual groups
Attendance management	Employees with attendance issues can be identified
Productivity improvement	Managers can develop strategies for improving performance